

QUESTIONS :

1. An ammeter is connected in----- with the load.
2. A voltmeter is connected in with the load.
3. Name two types of PMMC galvanometer meter movements: -----and-----
4. The same basic electrical principles govern the behavior of the PMMC meters and
5. The range of an ammeter, milliammeter, or microammeter can be increased if a ----- resistor is used in with the meter movement.
6. A voltmeter can be made from a dc current meter movement if a _____ resistor is connected in with the meter movement.
7. Draw the circuit diagram for a simple dc ohmmeter.
8. A current meter should have an internal resistance that is _____ compared with external circuit resistances . .
9. A voltmeter should have an internal resistance that is _____ compared with external circuit resistances.
10. An ohmmeter may be used in circuits with the power turned on. True or false?
11. A thermocouple ac ammeter reads the value of the ac waveform.
12. Thermocouple ammeters are used at frequencies up to MHz.
13. A hot-wire ammeter reads the value of the ac waveform.
14. A is an ac ammeter constructed from a movable coil in the magnetic fields of two stationary coils. All three coils are connected in _____ with each other.
15. The ammeter in Question 14 is calibrated in rms values but is actually reading - - - - - values.
16. A is used to extend the current range of the electro dynamometer type of ac current meter.
17. Iron-vane meters read the value of the ac waveform.
18. What happens to the iron-vane meter when dc is measured? Can you think of a method for overcoming this problem?
19. A rectifier meter reads the value of the alternating current or the ac voltage waveform.
20. A diode is often used as the instrumentation rectifier in audio frequency ac meters.
21. An ac meter reads different values depending on which way the meter probes are connected. This is called - - - - -
22. A rectifier-type ac voltmeter is operated such that the voltage applied to the rectifier diode is less than V_r This is an example of a _____ device.

23. Which type of ac meter rectifier is immune to both turnover effect and the effects of the phase of harmonics on the reading?

PROBLEMS

1. You have a 0-to-50- μ A dc current meter that has an internal resistance of 4500 Ω . What value of shunt resistor is needed to make the meter read 0 to 500 μ A full scale?
2. A dc current meter is in series with a 12- Ω load that is connected across a 9-V dc battery. How much current flows in the meter if it is assumed that both meter and battery are ideal?
3. A battery has an open-circuit (i.e., no-load) potential of 12 V dc and an internal resistance of 22 Ω . A 0-to-1-mA dc meter with an internal resistance is accidentally (!) connected across this battery. How much current flows in the meter movement?
4. Find the value of a shunt resistor required to make a 0-to-100- μ A dc current meter read 0 to 1 μ A if the meter's internal resistance is 1100 Ω . Use (a) Ohm's law and (b) current divider methods, and compare the results.
5. Solve Problem 4 for a 0-to-20- μ A meter movement.
6. What value of multiplier resistor will make a 0-to-100 μ A meter with an internal resistance of 1300 Ω read 0 to 100 V?
7. Calculate the multiplier resistance needed to make the meter in Problem 6 read 0 to 2 V. Show your work.
8. What is the sensitivity of a voltmeter constructed from a 0-to-30- μ A dc meter movement?
9. Calculate the impedance of a dc voltmeter on the 0-to-50-V scale if the sensitivity is 20,000 Ω /V.

CH2 QUESTIONS

1. Draw the circuit for a de Wheatstone bridge and label the components (R_1 , R_2 , R_3 , etc.).
2. Using the component designations given in Question 1, write the *null condition* equations.
3. Draw the circuit for a *Maxwell* bridge, label the components, and write the null condition equations.
4. Draw the circuit for a *Hay* bridge, label the components, and write the null condition equations.
5. Draw the circuit for a Schering bridge, label the components, and write the null condition equations.
6. What type of ac null detector is best suited for (a) the Wheatstone bridge, (b) the Hay bridge, (c) the Schering bridge, and (d) the Maxwell bridge?
7. What type of bridge is best suited for measuring capacitance?
8. What type of bridge is best suited for measuring the inductance of high- Q coils?
9. What type of bridge is best suited for measuring the inductance of low- Q coils?
10. What type of bridge is best suited for measuring the power factor of a capacitor?
11. A zero-center galvanometer with a range of 1-0-1 mA is used as the null detector in a de Wheatstone bridge. What type of circuit or device may be used to increase the sensitivity of the galvanometer so that 10 μ A will deflect the meter full scale?
12. What type of device or circuit may be used with an ac Wheatstone bridge to drive a ground-referenced null indicator?

Problems :

1. Is the Wheatstone bridge of Fig. 2-9 balanced? What is the value of E_0 ?
2. R_4 in Fig. 2-9 is changed to 5.2 Ω . What is the value of E_0 ?
3. A 20-k Ω resistor is connected between points A and B in Fig. 2-9, and R_4 is changed to 4.7 k Ω . Calculate the current through the 20-k Ω resistor.
4. What value would E_0 in Fig. 2-9 become if R_2 were shunted by a 100-k Ω resistor?

5. An unknown resistance R_x is connected into Fig. 2-9 in place of R_4 , and R_3 is adjusted to reestablish the null condition. What is the value of R_x if R_3 has a value of $576\ \Omega$?
6. What is the value of L_1 in Fig. 2-10?
7. What is the value of R_4 in Fig. 2-10?
8. Find the Q of the R_4/L_1 network in Fig. 2-10

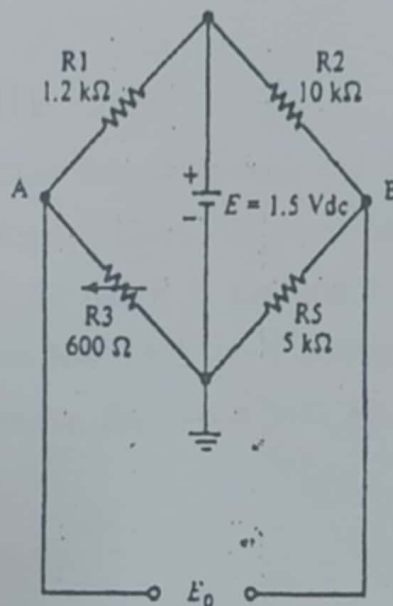


Fig. (2-9) Wheatstone bridge for Problems 1-5.

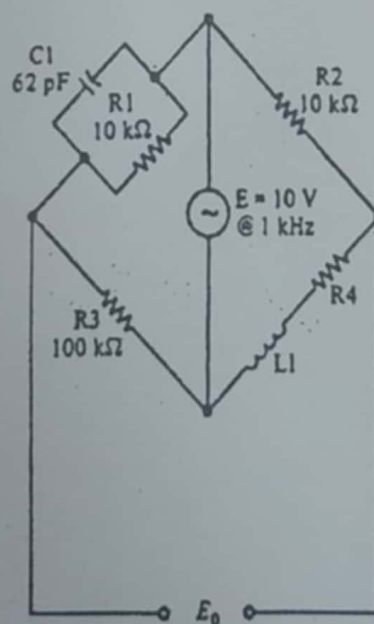


Fig. (2-10) Network for Problems 6-10.

9. A Hay bridge is formed by using the same components as in Fig. 2-10. An unknown inductor is connected in place of L_1 , and the bridge is adjusted to null. At that point it is found that $C_1 = 93 \text{ pF}$. What is the value of the unknown inductor?
10. What is the Q of the coil in Problem 9?
11. Use the loop method of network analysis to derive the expression for E_0 in an unbalanced de Wheatstone bridge.